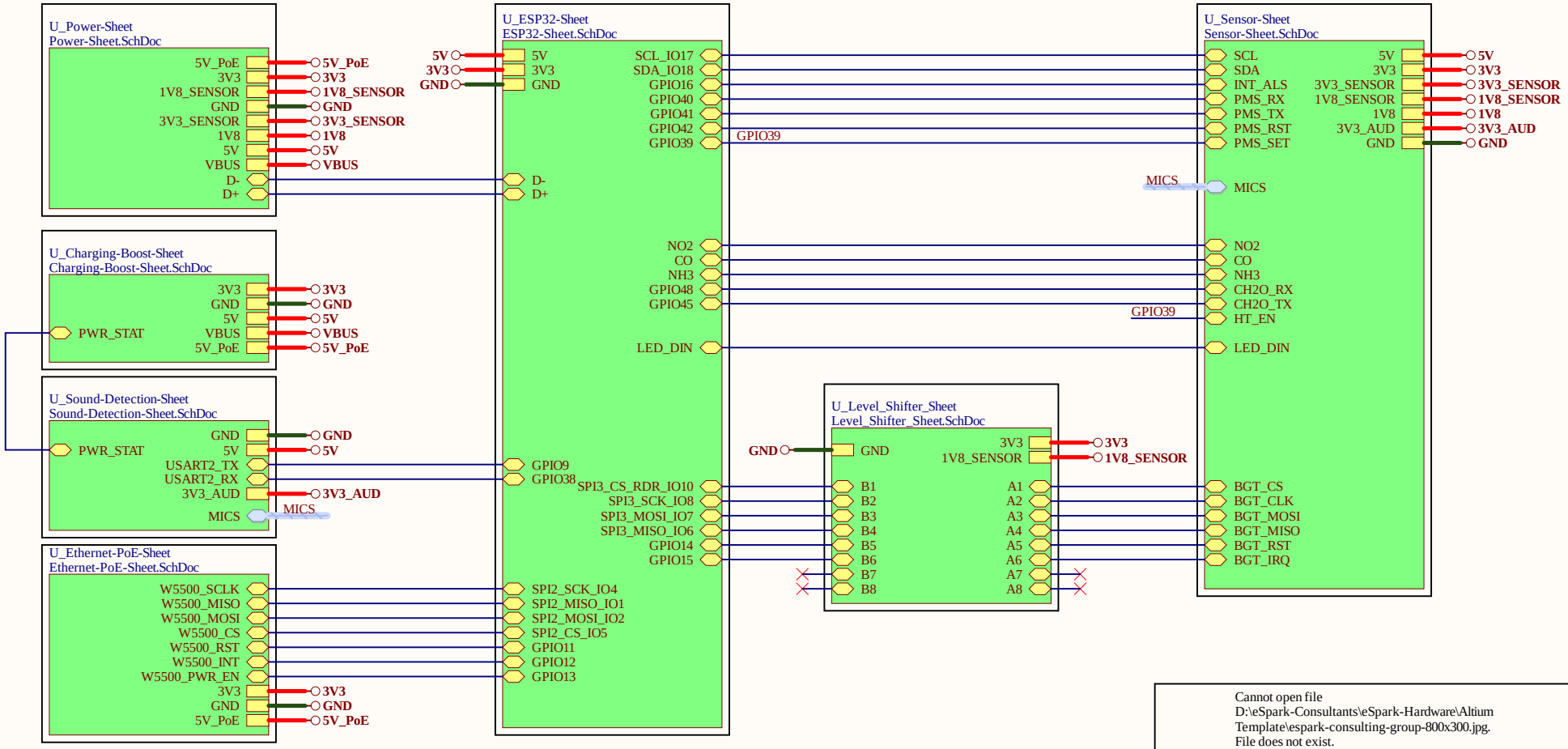
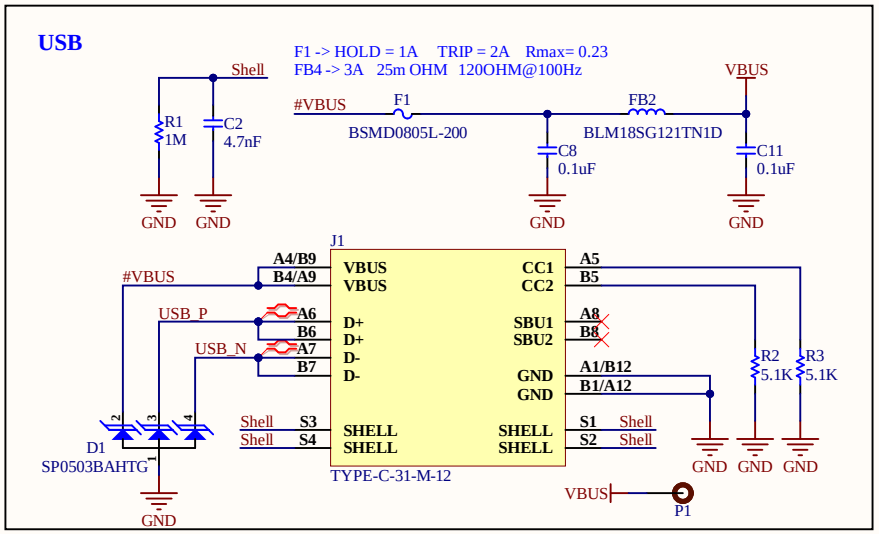
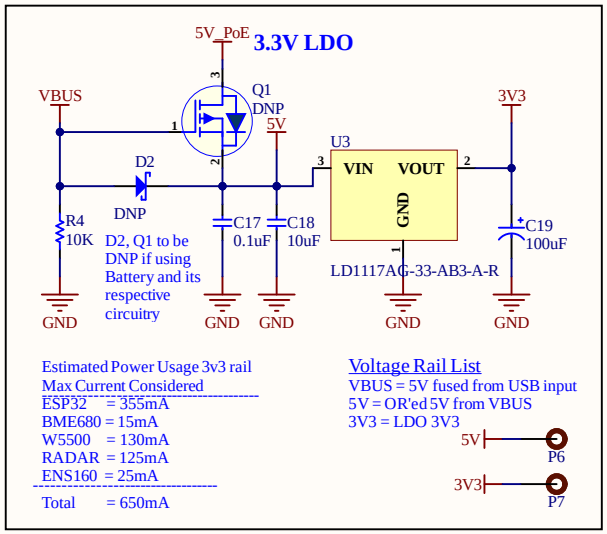
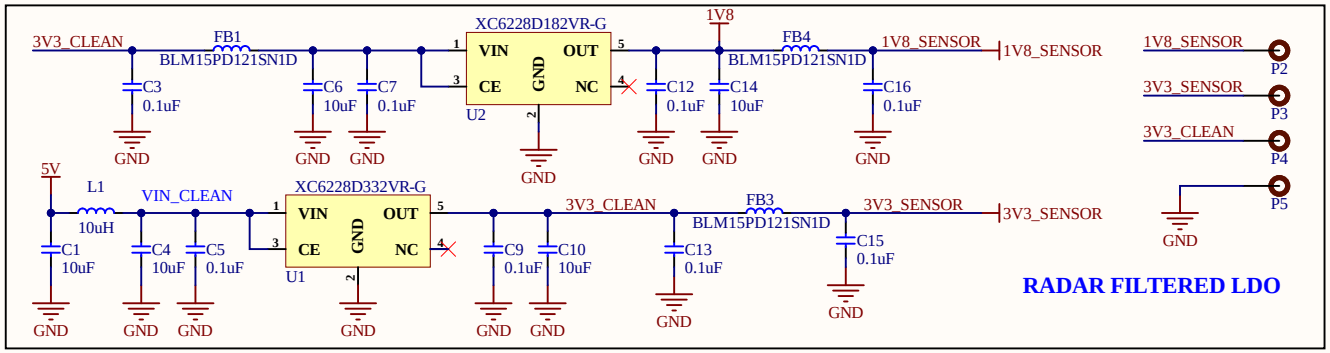


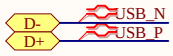
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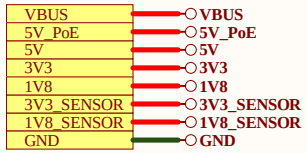
Title: Top-Sheet.SchDoc	
File: tracker-pro-airbnb-v1.0.PrjPcb	
Date: 7/14/2023	Page Num: 1
Drawn By: Huzaifa Ali	Page Total: 8









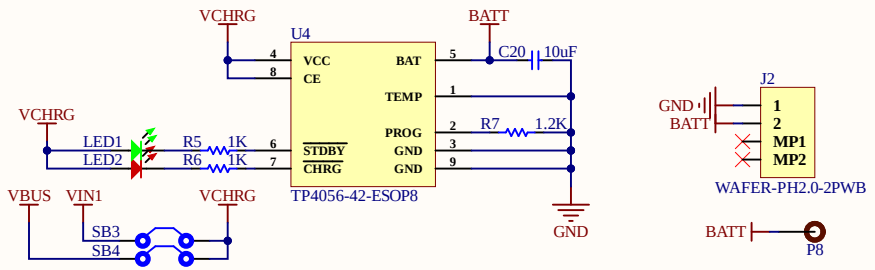


Power Section

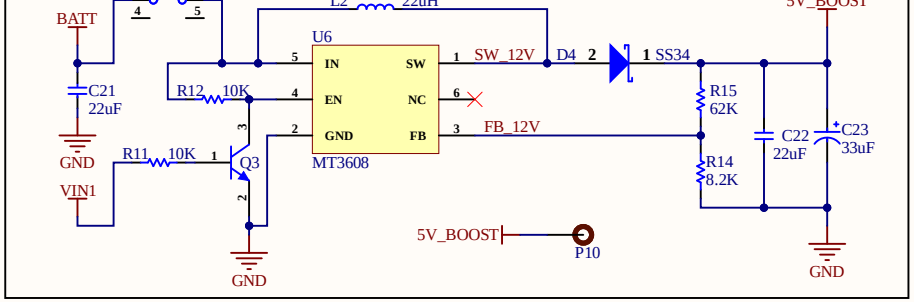
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Title: Power-Sheet.SchDoc	
File: tracker-pro-airbnb-v1.0.PrjPcb	
Date: 7/14/2023	Page Num: 2
Drawn By: Huzaifa Ali	Page Total: 8

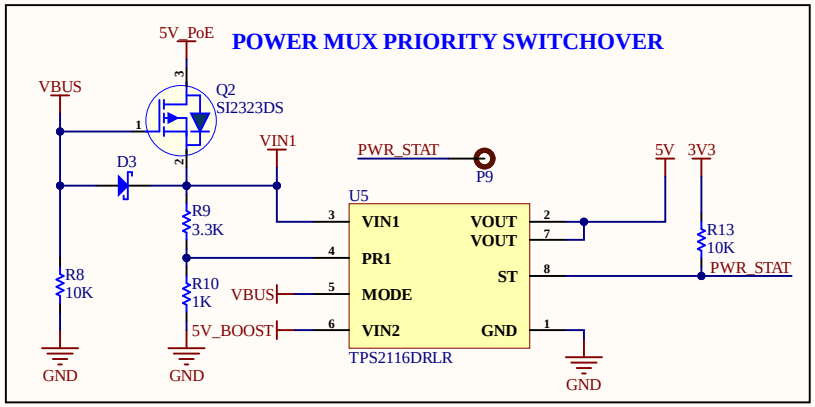
BATTERY CHARGING CIRCUIT



STEP UP CONVERTER



POWER MUX PRIORITY SWITCHOVER



PWR_STAT — PWR_STAT

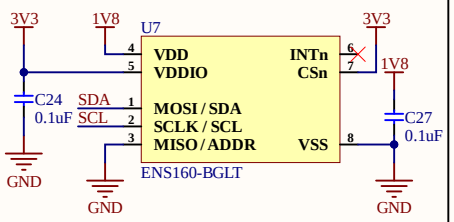
VBUS — VBUS
5V_PoE — 5V_PoE
5V — 5V
3V3 — 3V3
GND — GND

Charging Section

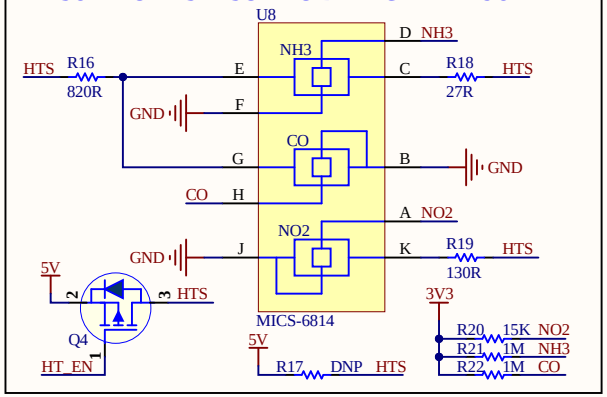
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Title: Charging-Boost-Sheet.SchDoc	
File: tracker-pro-airbnb-v1.0.PrjPcb	
Date: 7/14/2023	Page Num: 3
Drawn By: Huzaifa Ali	Page Total: 8

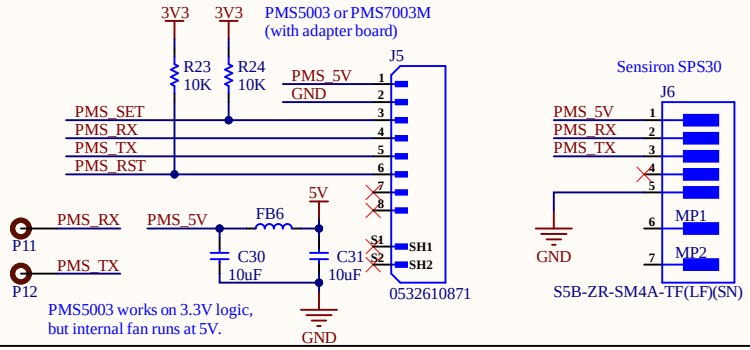
eCO2 TVOC AQI SENSOR



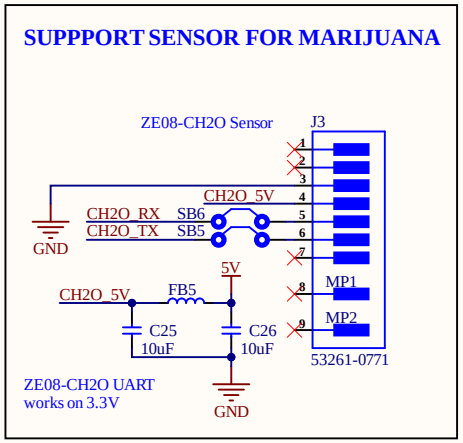
SUPPPORT SENSOR TO DETECT MARIJUANA



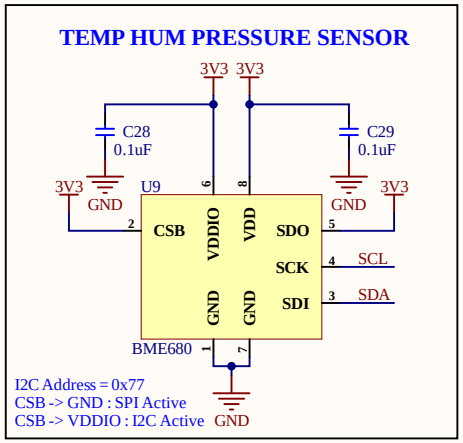
PM2.5 ENVIRONMENT SENSOR CIRCUIT



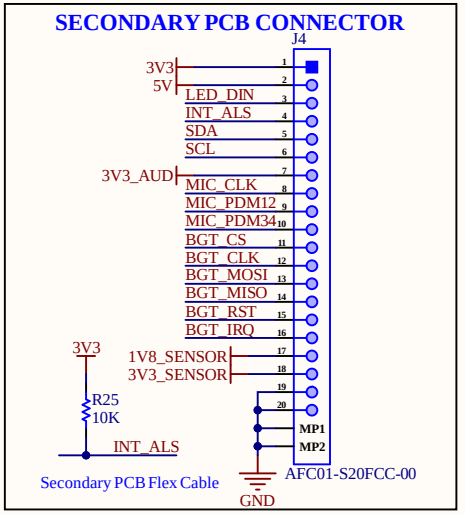
SUPPPORT SENSOR FOR MARIJUANA

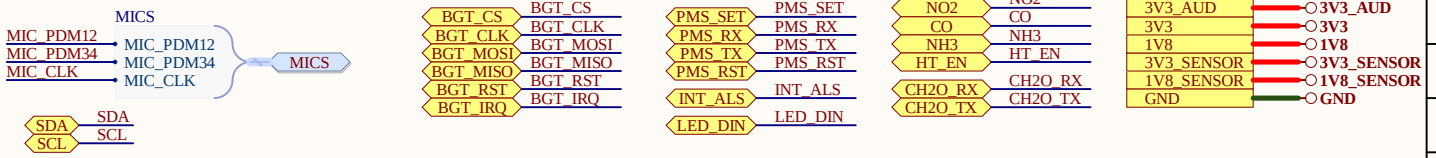


TEMP HUM PRESSURE SENSOR



SECONDARY PCB CONNECTOR



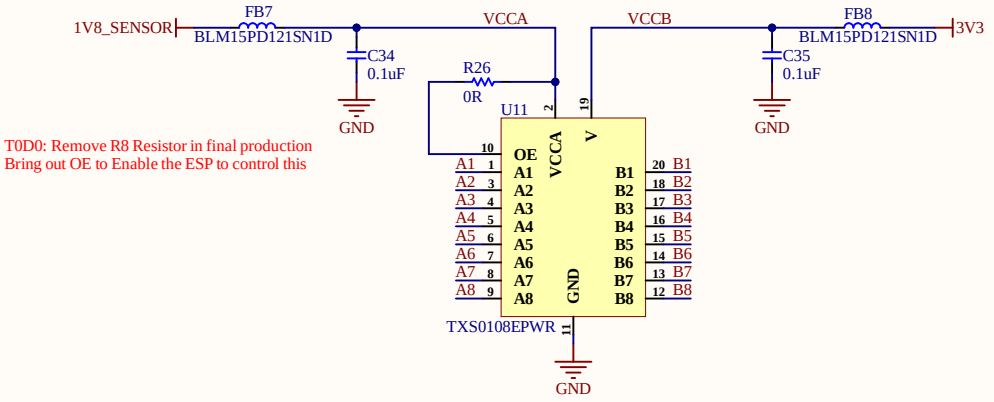


Sensors Section

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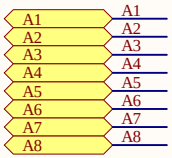
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Date: 7/14/2023	Page Num: 4
Drawn By: Huzaifa Ali	Page Total: 8

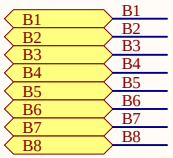
LEVEL SHIFTER

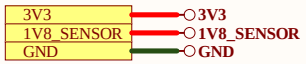


T0D0: Remove R8 Resistor in final production
Bring out OE to Enable the ESP to control this

1.4V to 3.6V on A-PORT and 1.65V to 5.5V on B-PORT(VCCA < VCCB)



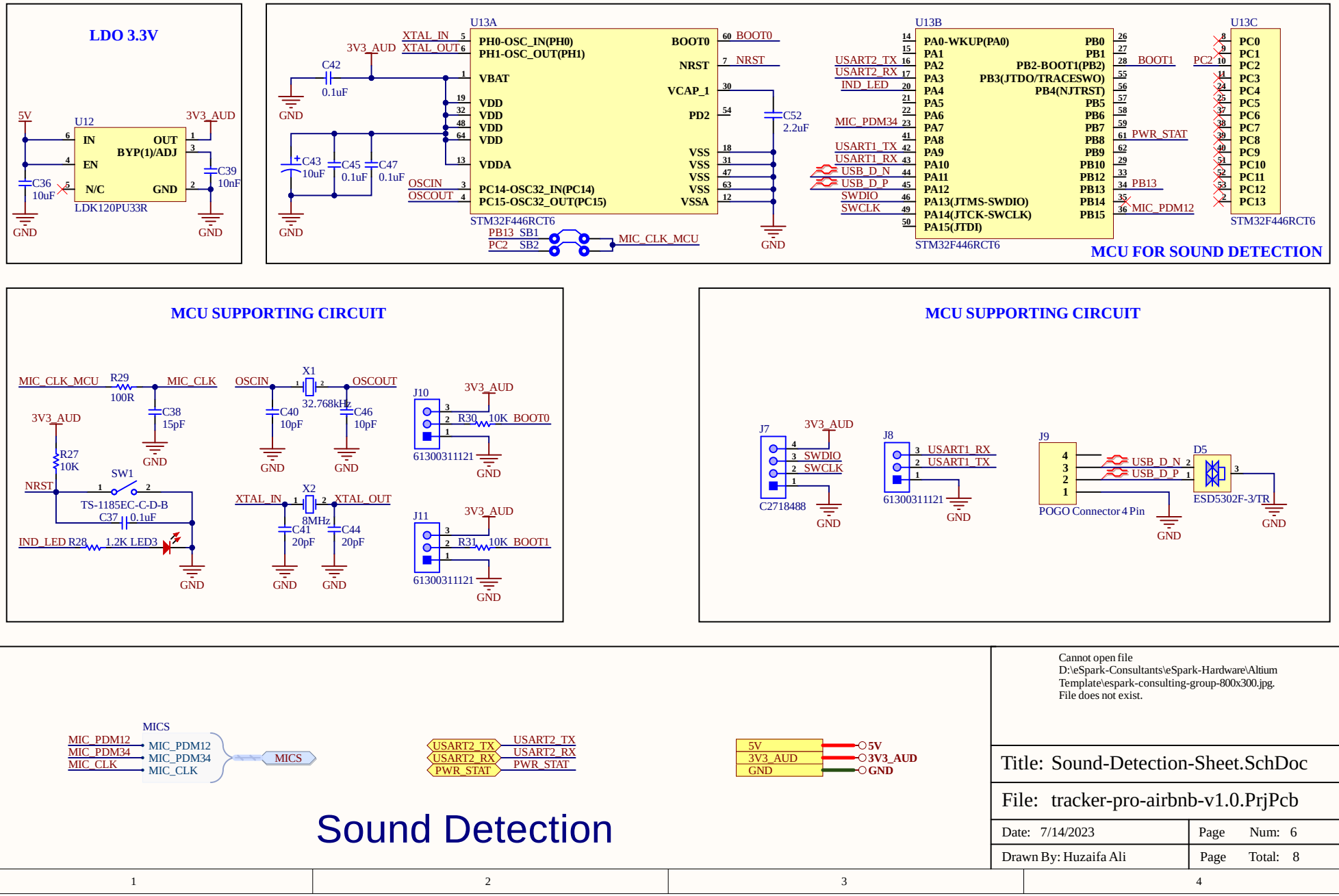




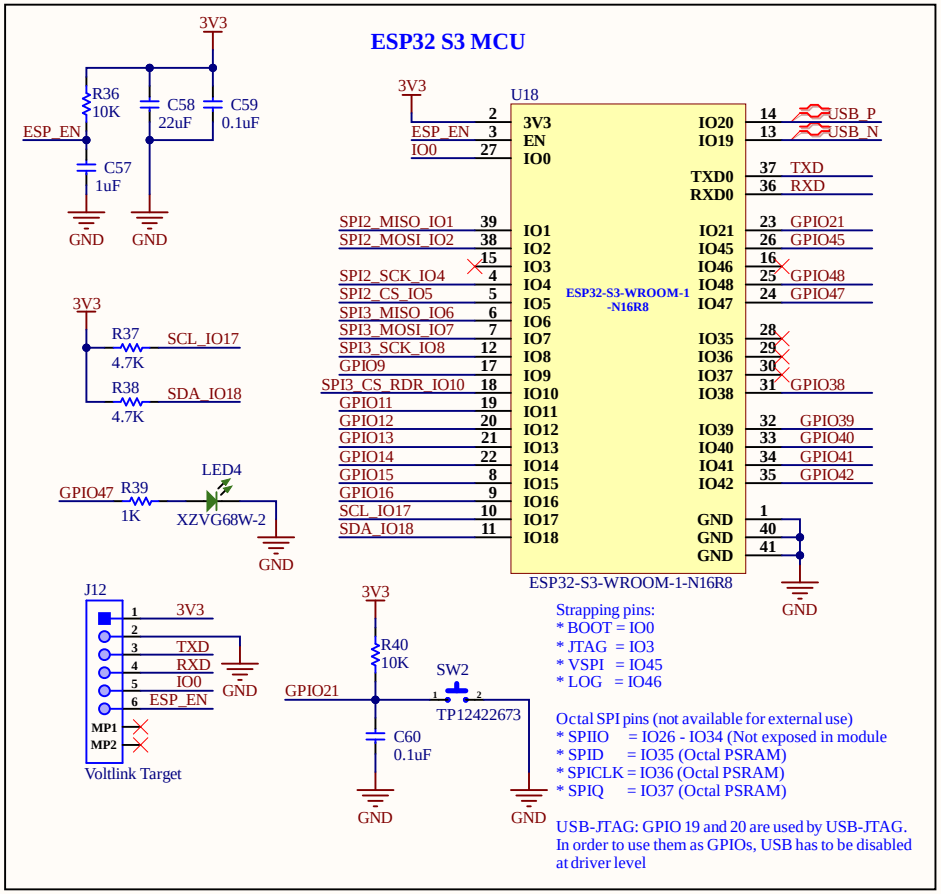
Level Shifter Section

Cannot open file D:\eSpark-Consultants\eSpark-Hardware\Altium Template\espark-consulting-group-800x300.jpg. File does not exist.

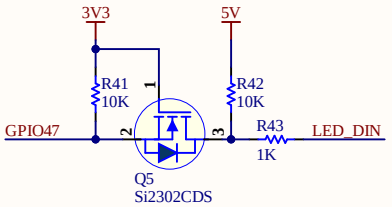
Title: Level_Shifter_Sheet.SchDoc	
File: tracker-pro-airbnb-v1.0.PrjPcb	
Date: 7/14/2023	Page Num: 5
Drawn By: Huzaifa Ali	Page Total: 8


LDO 3.3V
5V
U12
IN
OUT
BYP(1)/ADJ
EN
N/C
GND
3V3_AUD
C36
10uF
C39
10nF
LDK120PU33R
GND
GND
U13A
XTAL_IN
XTAL_OUT
3V3_AUD
C42
0.1uF
VBAT
VDD
VDD
VDD
VDD
VDDA
PC14-OSC32_IN(PC14)
PC15-OSC32_OUT(PC15)
PH0-OSC_IN(PH0)
PH1-OSC_OUT(PH1)
BOOT0
NRST
VCAP_1
PD2
VSS
VSS
VSS
VSS
VSSA
BOOT0
NRST
C52
2.2uF
C43
10uF
C45
0.1uF
C47
0.1uF
OSCIN
OSCOUT
STM32F446RCT6
PB13 SB1
PC2 SB2
MIC_CLK_MCU
U13B
PA0-WKUP(PA0)
PA1
PA2
PA3
PA4
PA5
PA6
PA7
PA8
PA9
PA10
PA11
PA12
PA13(JTMS-SWDIO)
PA14(JTCK-SWCLK)
PA15(JTDI)
PB0
PB1
PB2-BOOT1(PB2)
PB3(JTDO/TRACESWO)
PB4(NJTRST)
PB5
PB6
PB7
PB8
PB9
PB10
PB12
PB13
PB14
PB15
USART2_TX
USART2_RX
IND_LED
MIC_PDM34
USART1_TX
USART1_RX
USB_D_N
USB_D_P
SWDIO
SWCLK
BOOT1
PWR_STAT
PB13
MIC_PDM12
STM32F446RCT6
U13C
PC0
PC1
PC2
PC3
PC4
PC5
PC6
PC7
PC8
PC9
PC10
PC11
PC12
PC13
PC2
STM32F446RCT6
MCU FOR SOUND DETECTION
MCU SUPPORTING CIRCUIT
MIC_CLK_MCU
R29
100R
MIC_CLK
C38
15pF
3V3_AUD
R27
10K
SW1
NRST
TS-1185EC-C-D-B
C37
0.1uF
IND_LED
R28
1.2K
LED3
GND
OSCIN
X1
32.768kHz
OSCOUT
C40
10pF
C46
10pF
XTAL_IN
X2
8MHz
XTAL_OUT
C41
20pF
C44
20pF
J10
3V3_AUD
R30
10K
BOOT0
61300311121
J11
3V3_AUD
R31
10K
BOOT1
61300311121
MCU SUPPORTING CIRCUIT
J7
3V3_AUD
SWDIO
SWCLK
C2718488
J8
USART1_RX
USART1_TX
61300311121
J9
USB_D_N
USB_D_P
POGO Connector 4 Pin
D5
ESD5302F-3/TR
MICS
MIC_PDM12
MIC_PDM34
MIC_CLK
USART2_TX
USART2_RX
PWR_STAT
5V
3V3_AUD
GND
5V
3V3_AUD
GND
Sound Detection
Cannot open file D:\eSpark-Consultants\eSpark-Hardware\Altium Template\espark-consulting-group-800x300.jpg. File does not exist.
Title: Sound-Detection-Sheet.SchDoc
File: tracker-pro-airbnb-v1.0.PrjPcb
Date: 7/14/2023
Page Num: 6
Drawn By: Huzaifa Ali
Page Total: 8

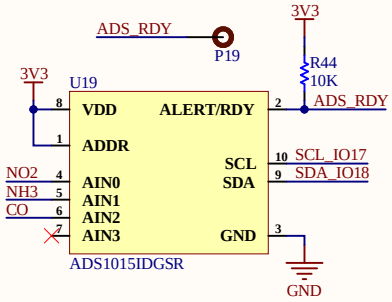
ESP32 S3 MCU

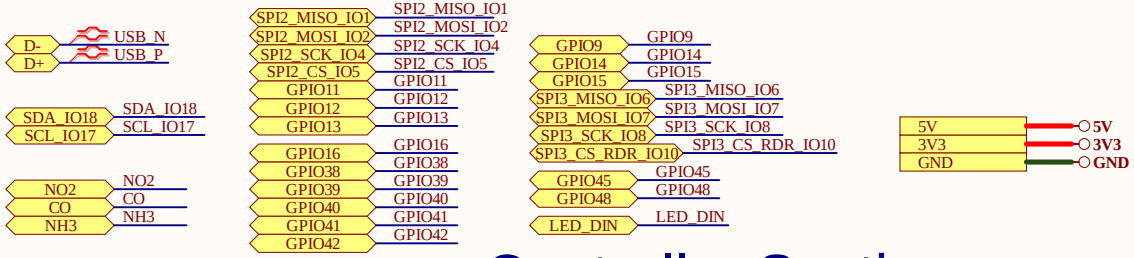


RGB LEDs LEVEL SHIFTER



12-BIT ADC FOR MICS-6814 SUPPORT



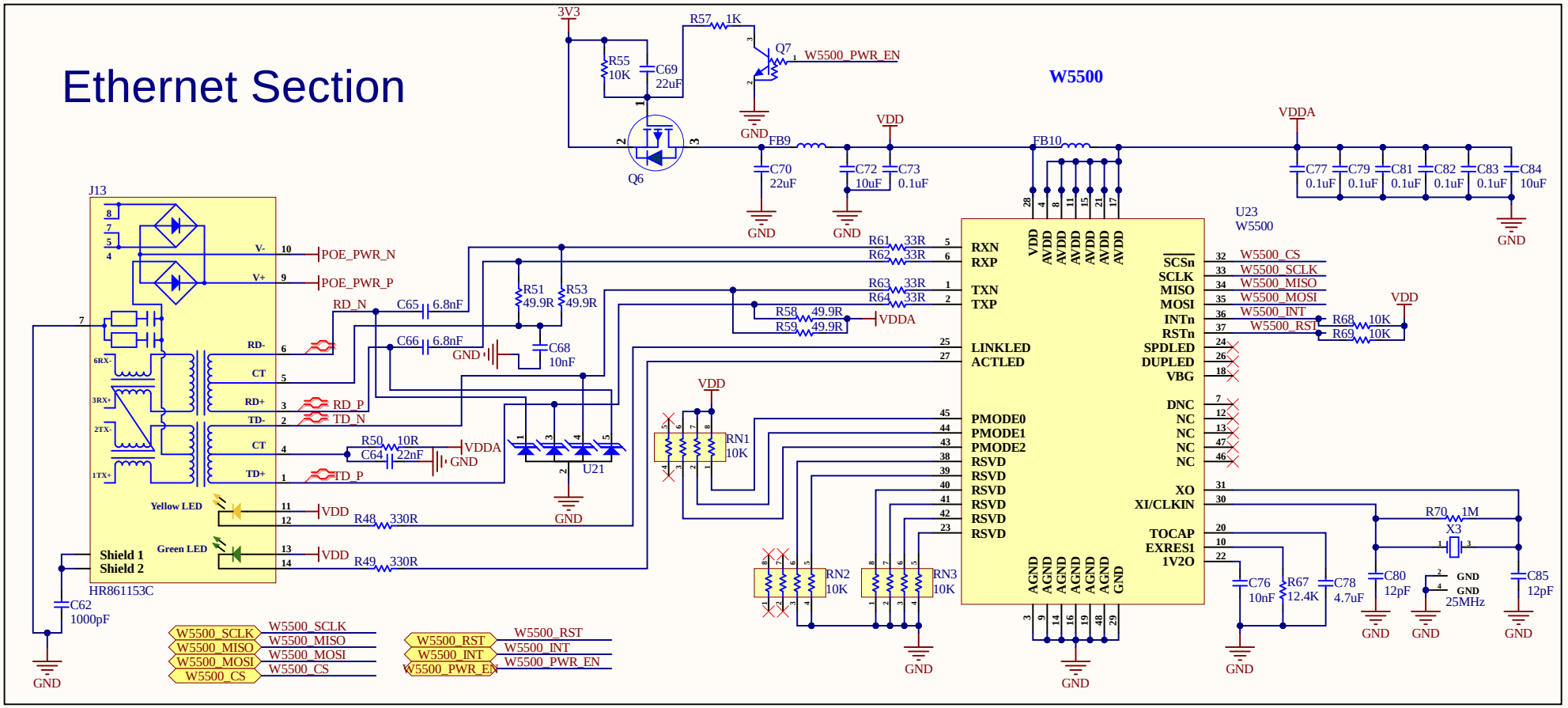


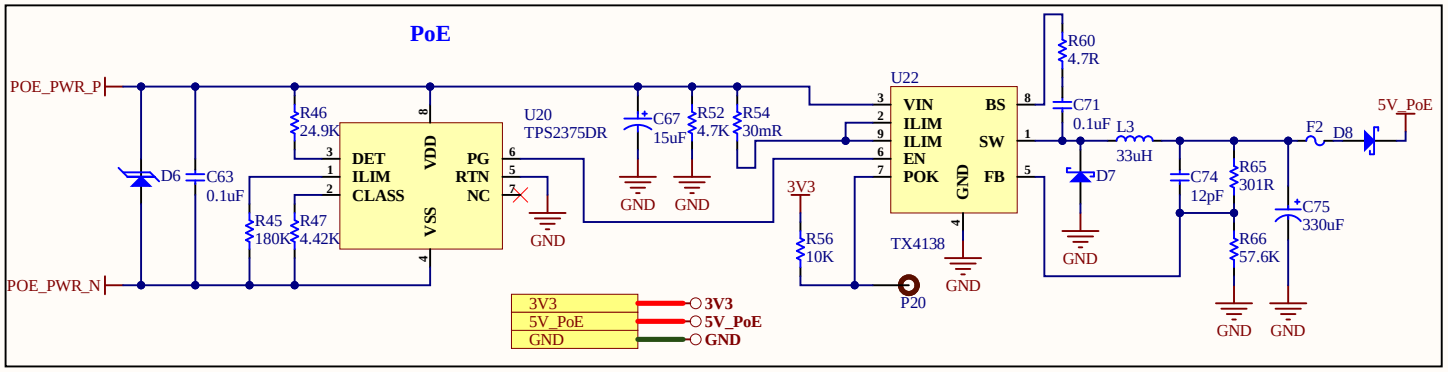
Controller Section

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Title: ESP32-Sheet.SchDoc	
File: tracker-pro-airbnb-v1.0.PrjPcb	
Date: 7/14/2023	Page Num: 7
Drawn By: Huzaifa Ali	Page Total: 8

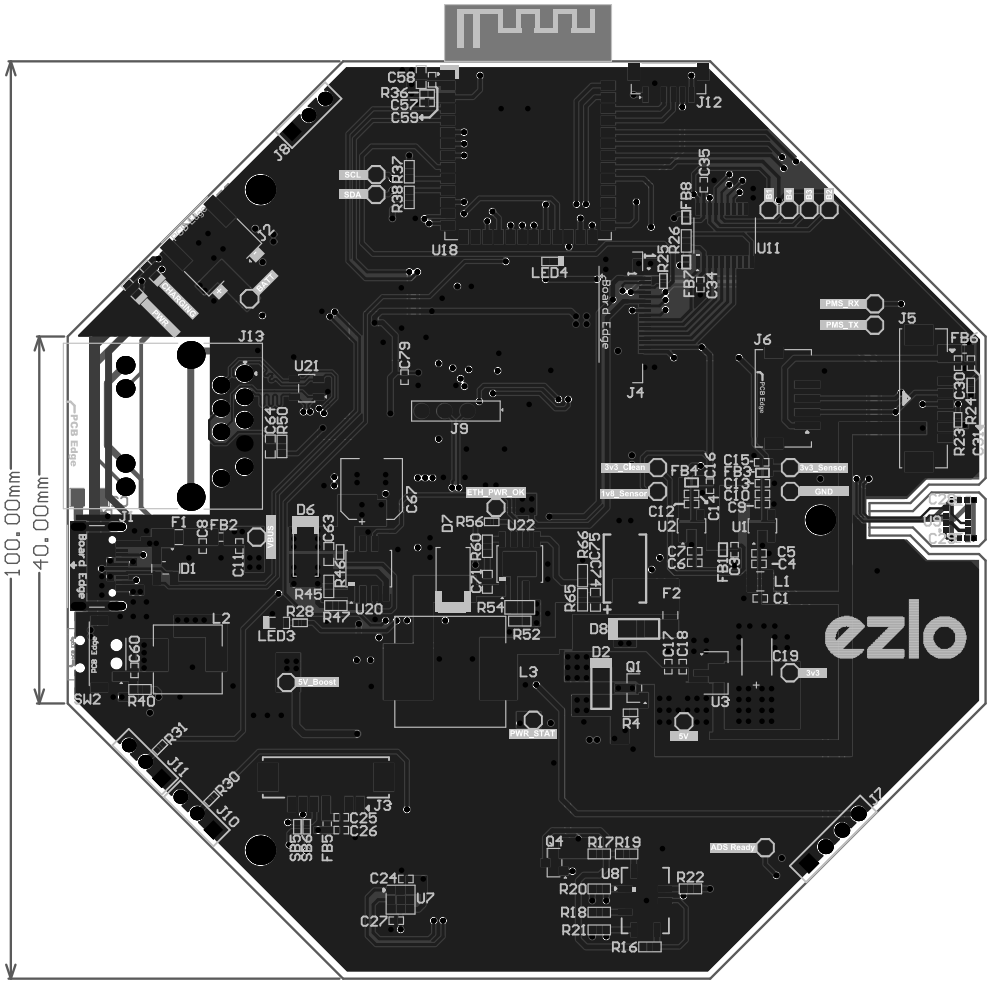
Ethernet Section





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File does not exist.

Title: Ethernet-PoE-Sheet.SchDoc	
File: tracker-pro-airbnb-v1.0.PrjPcb	
Date: 7/14/2023	Page Num: 8
Drawn By: Huzaifa Ali	Page Total: 8



Layer	Name	Material	Thickness	Constant	Board Layer Stack
	Top Overlay				
	Top Solder	Solder Resist	0.39mil	3.3	
1	Top Layer		1.38mil		
	Dielectric 2	FR-4	8.27mil	4.6	
2	Signal + GND	CF-004	0.60mil		
	Dielectric 1	FR-4	41.93mil	4.6	
3	Signal + 3v3	CF-004	0.60mil		
	Dielectric 3	FR-4	8.27mil	4.6	
4	Bottom Layer		1.38mil		
	Bottom Solder	Solder Resist	0.39mil	3.3	
	Bottom Overlay				
Total board thickness:			63.21mil		